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# ICPC Algorithm Library

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# 目录

|     |          |   |
|-----|----------|---|
| 1   | Geometry | 1 |
| 1.1 | 点和向量     | 1 |
| 1.2 | 线段和多边形   | 1 |

# 1 Geometry

## 1.1 点和向量

```
using db = long double;
struct p2 {
    db x, y;
    db abs() { return hypot(x, y); }
    p2 operator+(p2 b) { return {x + b.x, y + b.y}; }
    p2 operator-(p2 b) { return {x - b.x, y - b.y}; }
    p2 operator*(db k) { return {x * k, y * k}; }
    p2 operator/(db k) { return {x / k, y / k}; }
    db operator*(p2 b) { return x * b.x + y * b.y; } // 点积
    db operator%(p2 b) { return x * b.y - y * b.x; } // 叉积
};
using ps = vector<p2>;
```

## 1.2 线段和多边形

```
db toLine(p2 p, p2 a, p2 b) { return abs((b - a) % (p - a)) / (b - a).abs(); }
```